

Seven gene sets. One auditable curation.

A concise record of the publication rules, pathway selection, identity checks, and overlap that produced the stocker-ready inputs.

7

stocker CSVs

295

unique FBgns

4

publication sets

3

pathway sets

The final library at a glance

Four evidence-defined sets and three pathway-hit sets; 295 unique genes after FBgn deduplication.

01

6

genes

Mechanistic

Publication core + screen
ranks 1–6

02

20

genes

Homeostatic H×R

15 History+/Rebound– + 5
opposite

03

97

genes

CSW 4+

Wake frequency ≥ 4 ,
threshold-scoped

04

4

genes

HLH regulators

Four bHLH factors named in
paper

05

99

genes

Ribosomal

DAVID term hits; approved
keywords

06

167

genes

Mitochondrial

DAVID mito/metabolism term
hits

07

5

genes

Immune

DAVID immune / Toll / Imd
term hits

 Σ

295

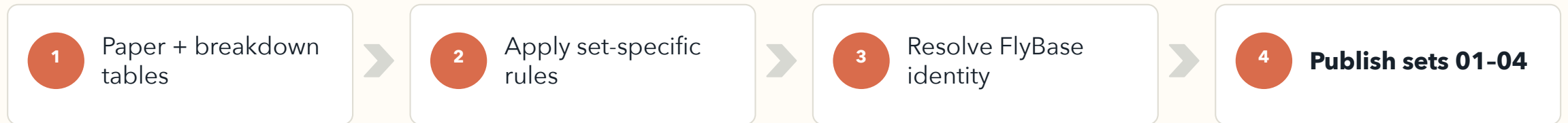
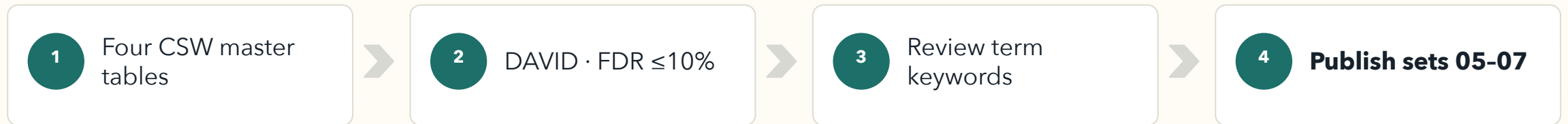
unique FBgns

Seven-set union

Deduplicated on
flybase_gene_id

Two curation tracks converge on one controlled folder

The shared gates are identity resolution, FBgn deduplication, and a clean seven-file handoff.

TRACK A**Publication categories****TRACK B****Pathway hits****902**

source observations

791

unique source FBgns

≥2

minimum pathway frequency

7

experiments

Thresholds remain separate: FC0.5 and FC1 are never collapsed into a max-frequency field.

Publication sets use explicit, reproducible rules

Sets 01-02 preserve publication evidence and resolve one biologically important identity conflict.

SET 01 · 6 GENES

Mechanistic

Publication core joined by FBgn to mechanistic screen ranks 1-6.

- Carry forward mechanism, literature status, Tx-Omics evidence, and fl.ai confidence.
- Evidence labels: consistent, correlated, or curated.
- No rank or annotation is inferred outside the source tables.

unc79 · SIFa · rumpel · AstA-R2 · Trhn · RpL23

SET 02 · 20 GENES

Homeostatic history × rebound

15

History+ / Rebound-

5

History- / Rebound+

Identity correction

Source row trh / FBgn0262139 is resolved from publication context to Trhn / FBgn0035187 (tryptophan hydroxylase). The wrong FBgn is never retained.

Screen frequency and direct-regulator evidence stay separate

Set 03 is threshold-scoped screen evidence; Set 04 comes directly from the paper.

SET 03 · 97 GENES

CSW 4+ wake-frequency union

97

FC0.5 wake frequency 4-6

7

also qualify at FC1

0

sleep genes at frequency 4

Union rule

- Take FC0.5 wake genes at frequency 4, 5, or 6.
- Add FC1 wake frequency-4 genes; all seven are already in the FC0.5 set.
- Keep FC0.5 and FC1 metadata in separate columns.

SET 04 · 4 GENES

HLH regulators

Named in Results / Table S1 after HOMER identified a CAGCTG E-box upstream of wake-induced genes.

bigmax FBgn0039509

HLH3B FBgn0011276

E(spl)m3-HLH FBgn0002609

E(spl)mbeta-HLH FBgn0002733

Pathway sets contain DAVID hits—not every annotated gene

Only hit genes from approved FDR-passing terms enter sets 05-07.



Mitochondrial / metabolism



167

Ribosomal / translation



99

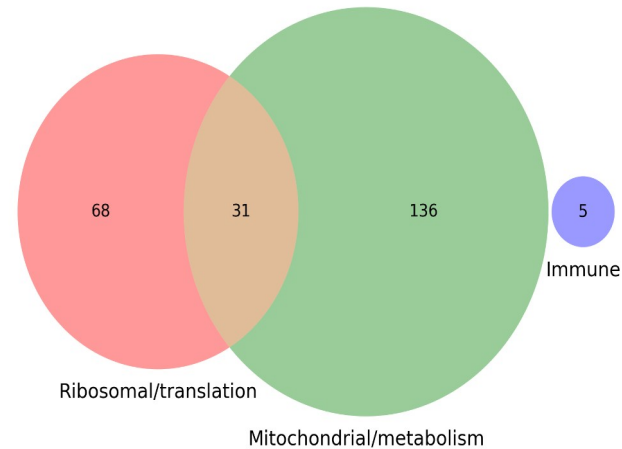
Immune



5

All three lists are wake-only in this run.

Pathway gene overlap



Why the overlap is intentional

Mitochondrial translation and mitochondrial ribosomal-subunit terms match both keyword groups. Their hit genes are retained in every matched bucket.

31 **ribo** $\cap$ **mito**

Identity resolution is conservative by design

Current symbols can pass directly; ambiguous synonyms require biological context or a case-supported source match.

PASS**Trhn****FBgn0035187**

Current symbol · tryptophan hydroxylase neuronal · publication-resolved in Homeostatic.

REJECT**trh****FBgn0262139**

Current symbol · trachealess · must not enter any generated identity field for this publication row.

REVIEW**Trh****ambiguous**

Synonym of Hn, Trhn, and trh · never auto-approved without biological context.

Approval gates

- Exact current symbol
- Publication-supported correction
- Unique case-insensitive current-symbol match
- Otherwise: stop for review

Membership key

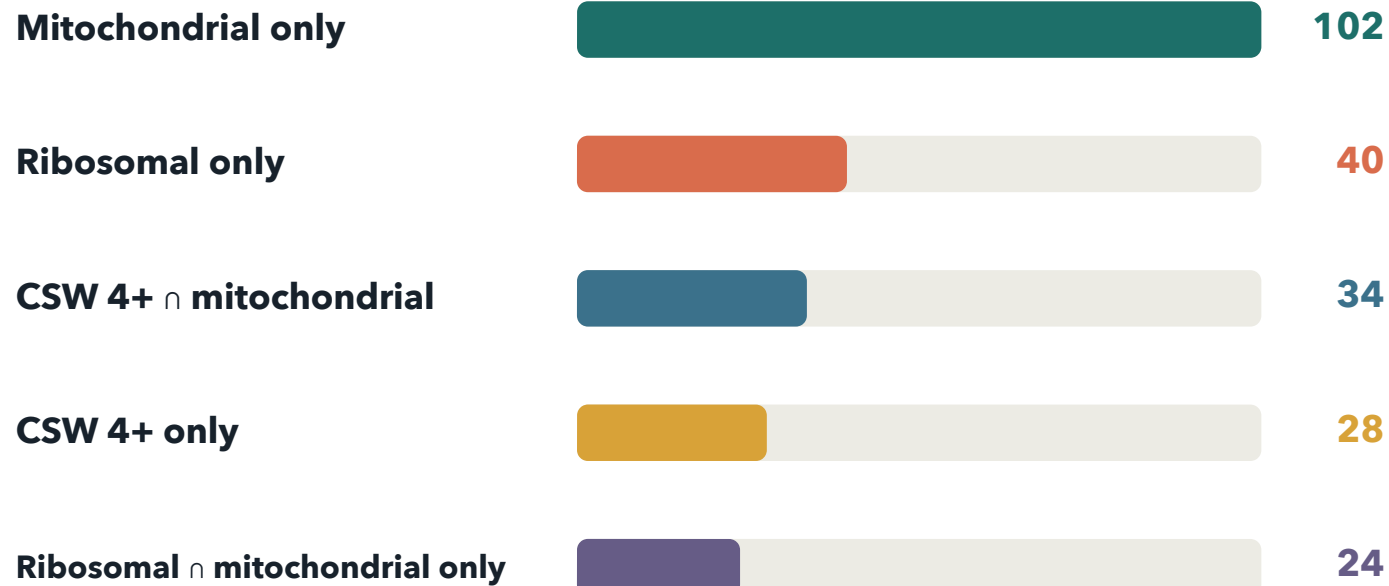
flybase_gene_id

Symbols are display labels. All overlap, deduplication, and set membership use FBgn only.

Most genes are set-specific; overlap concentrates in CSW and pathways

Exact intersections expose the reusable biological core without inflating membership.

Largest exact intersection blocks



The immune and HLH sets remain fully distinct from the pathway core.

What matters

200 genes in exactly one set

87 genes in exactly two sets

8 genes in exactly three sets

Largest pairwise overlaps

41 CSW 4+ $\cap$ mito

35 CSW 4+ $\cap$ ribo

31 ribo $\cap$ mito

The contract is simple—and strict.

7

Exactly seven CSVs
in the discovery
folder

2

Required keys:
ext_gene and
flybase_gene_id

0

No overlap, GO, or
audit CSVs beside
the inputs

RUN

```
python -m fl_ai_reagent_stocker run ./data/gene_sets/Tx-Omics_Revision/ \
--config ./data/config/stock_split_config_priority_example.json
```

Seven inputs

- 01 Mechanistic
- 02 Homeostatic HxR
- 03 CSW 4+
- 04 HLH regulators
- 05 Ribosomal
- 06 Mitochondrial
- 07 Immune

Stability rule

Discovery is recursive: every extra *.csv changes the run. Keep analysis outputs elsewhere.